

JIAJIE LI

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SUMMARY

Ph.D. candidate in Computer Science focusing on LLM domain adaptation and evaluation. Experienced in developing scalable methods for landing LLMs in specialized domains (e.g., surgery) and designing LLM Judge systems for robust downstream evaluation. Skilled in bridging theory and practice to deploy and assess LLMs in real-world applications.

EXPERIENCE

Bytedance, Research Engineer Intern, AI Foundation Team Jan 2025 – July 2025

- Built an adaptive **LLM-as-a-Judge** framework for LLM evaluation service that learns task-specific evaluation dimensions from failure cases, boosting reward model accuracy of proprietary LLM by **3.3%** on RewardBench
- Proposed a standard operation procedure (**SOP**) for structured LLM evaluation that automatically analyzes the output structures and selects appropriate structure- and field-wise evaluation tools via **LLM tool use**
- Designed an **agent-based** LLM-as-a-Judge method for end-to-end evaluation of structured LLM outputs
- Built a modular library of **MCP tools** enabling LLM evaluation and analysis of structured outputs with tool use
- Skills: Python, Shell, Git, PyTorch, LLM, Deep Learning, LangChain, MCP, vLLM, Prompt Engineering

Motorola Solutions, Research Intern, Embedded Video AI Team May 2023 – Sept 2023

- Implemented a **soft-prompt tuning**-based object detection system that quickly adapts to new tasks with minimal training cost, outperforming the in-production model on 8 of 9 tasks using only **0.1% of the training data**
- Developed a few-shot object detection model that uses reference images as prompts, enabling rapid adaptation to new tasks without fine-tuning and reducing adaptation time **from hours to milliseconds**
- Skills: Python, PyTorch, Deep Learning, Object Detection, Prompt Tuning, Slurm

IBM Research, Research Intern, Hybrid Cloud Team |  [Github](#) May 2022 – Sept 2022

- Designed and implemented **fault-tolerant** and **elastic** distributed training capabilities for large-scale foundation models on **Ray** Cluster using **TorchX**, improving system reliability under dynamic cluster conditions
- Rebuilt TorchX's Ray node lifecycle management mechanism by integrating an async FSM for robust scaling
- Skills: Python, Git, Distributed Training, Deep Learning, Machine Learning, Ray, Horovod, TorchX, K8s

EDUCATION

University at Buffalo, Ph.D. in Computer Science and Engineering, advisor: Jinjun Xiong Feb 2023 – Dec 2026

University at Buffalo, M.S. in Computer Science and Engineering Sept 2020 – Dec 2022

Hangzhou Dianzi University, B.S. in Information and Computing Science Sept 2015 – Jun 2019

SELECTED PROJECTS

RASO, Open-set Surgical Object Recognition Model | Accepted by ICLR'25 |  [GitHub](#) June 2024 – Sept 2024

- Developed RASO, a foundation model for recognizing any surgical object, achieving state-of-the-art zero-shot and supervised recognition performance across surgical benchmarks by leveraging only weakly-supervised data
- Designed a scalable data generation pipeline, processing **2,201** surgical procedures and generating **3.6 million** tag annotations without human intervention, significantly reducing the need for manual annotations
- Achieved a **10.6 mAP improvement (18%)** over SOTA models in zero-shot recognition tasks on public benchmarks

LLaVA-Surg, Vision Language Model for Surgical Videos Feb 2024 – May 2024

- Proposed Surg-QA, the first large-scale instruction-tuning dataset for surgical videos of **102K video Q&A pairs**
- Developed LLaVA-Surg, the first expert-level video conversational model for multimodal surgical understanding, achieving **85.6%** improvement over prior state-of-the-art multimodal LLMs on our surgical VQA benchmark

xMLP, Quadratic Neural Network for Private Inference |  [Github](#) May 2022 – March 2023

- Proposed an arithmetic-only neural network architecture for computer vision tasks, achieving **+2.3% higher accuracy** than ResNet with **80% fewer parameters** and up to **47× faster private inference speed**
- Developed an optimized private inference framework, delivering **6× faster linear-layer inference speed** and **19% faster nonlinear-layer inference speed** compared to the state-of-the-art private inference framework CrypTen

SELECTED PUBLICATIONS

[ICLR'25 Spotlight (5.1%)] 1st author. *Recognize Any Surgical Object: Unleashing the Power of Weakly-Supervised Data*

[ICCV'23] 2nd author. *Extensible and efficient proxy for neural architecture search*

[NeurIPS'23] 2nd author. *SyncTREE: fast timing analysis for integrated circuit design through a physics-informed tree-based graph neural network*